

AIGR Deep Research Report: ABHD8 (Q96I13) — Lipid Catalysis versus Inflammasome Adaptor

Gene: ABHD8 | **Organism:** *Homo sapiens* (NCBITaxon:9606) | **UniProt:** Q96I13 **Focus type:** function_assignment | **Hypothesis slug:** lipid-catalysis-versus-inflammasome-adaptor **Source:** `genes/human/ABHD8/ABHD8-ai-review.yaml` (free-text selector)

Summary

The seed hypothesis proposes that human ABHD8 retains one or more specific lipid functions — glycerophospholipase activity, generic ester hydrolase activity, lysophosphatidic acid (LPA) acyltransferase activity, participation in phosphatidic acid biosynthesis, or lipid homeostasis — and asks that each be adjudicated separately from ABHD8's demonstrated NLRP3 adaptor role. It further argues, correctly as a methodological principle, that neither the presence of an adaptor function nor the absence of substrate experiments by itself establishes catalytic or pathway loss. After three iterations of sequence, domain, orthology, and literature analysis, the evidence resolves into a clean split between **catalytic *potential* (supported)** and **specific lipid *activities* (unsupported by direct evidence, and in one case mechanistically implausible)**.

ABHD8 possesses a fully intact, canonical serine-hydrolase catalytic triad — **Ser252 (nucleophile), Asp370 (acid), His398 (base)** — embedded in a classic GX SXG nucleophile elbow (the motif **GHSYG**) within a complete α/β -hydrolase (AB_hydrolase-1) domain. This triad is conserved across all reviewed vertebrate orthologs from human to *Xenopus*, spanning more than 360 million years of divergence, which indicates purifying selection to preserve catalytic competence. ABHD8 is therefore **not** a dead pseudoenzyme, and a blanket claim of catalytic or pathway loss would be unjustified. This directly vindicates the seed's caution.

At the same time, **no ABHD8-specific biochemical assay demonstrates any lipid substrate**. The specific lipid GO terms are phylogenetic/family inferences (IBA from GO_Central; IEA from InterPro), not experiments. The LPA-acyltransferase claim is mechanistically inconsistent with

a Ser-His-Asp hydrolase fold and almost certainly reflects paralog/family carry-over. The **only** experimentally demonstrated function is the NLRP3 scaffold/adaptor role: ABHD8 recruits the palmitoyltransferase ZDHHC12 to NLRP3, promoting NLRP3 palmitoylation and chaperone-mediated-autophagy (CMA) degradation (PMID: 39225180). Accordingly, curation should retain the NLRP3 term (GO:1900226, IMP) and cytoplasm (GO:0005737, IDA) as core, keep any lipid-catalysis annotation at most as generic hydrolase potential (GO:0003824, IEA), and treat the specific lipid terms as over-annotations pending direct substrate assays.

Verdict: Partially supported / unresolved — leaning over-annotated for the specific lipid activities, while catalytic potential is genuinely supported.

Key Findings

Finding 1 — ABHD8 has an intact canonical serine-hydrolase catalytic triad but no demonstrated substrate

UniProt Q96I13 annotates an **AB_hydrolase-1 domain** spanning residues ~177–279, with the nucleophile-elbow **GXSXG** motif present as **G250-H-S252-Y-G254 (GHSYG)**. The charge-relay / catalytic triad is annotated at **Ser252 (nucleophile)**, **Asp370 (acid)**, **His398 (base)**, and the residue identities S/D/H were confirmed directly from the sequence. The protein maps to **InterPro IPR000073 (AB_hydrolase_1)**, **Pfam PF00561**, and the ABHD8-specific PANTHER subfamily **PTHR42886:SF83**.

Critically, UniProt carries **no CATALYTIC ACTIVITY comment and no specific reaction** for ABHD8. The only **FUNCTION** comment is the NLRP3 scaffold role citing

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The molecular-function GO annotation is the generic **GO:0003824 (catalytic activity)** with evidence **IEA:InterPro** only. In other words, the domain architecture supports catalytic *capability*, but nothing in the primary record pins down what ABHD8 actually hydrolyzes. The presence of a textbook triad establishes potential; it does not establish any specific reaction. This is the crux of the adjudication: catalytic potential and demonstrated function are, at present, decoupled.

Finding 2 — The specific lipid-pathway GO terms are phylogenetic/family inference, not experiment

The GO annotation profile for Q96I13 separates cleanly into "inferred" versus "experimental":

GO term	Aspect	Evidence code	Status
GO:0006654 phosphatidic acid biosynthetic process	BP	IBA:GO_Central	Phylogenetic inference — no experimental support
GO:0003824 catalytic activity	MF	IEA:InterPro	Generic, electronic
GO:1900226 negative regulation of NLRP3 inflammasome complex assembly	BP	IMP:UniProtKB	Experimental
GO:0005737 cytoplasm	CC	IDA	Experimental

The "phosphatidic acid biosynthetic process" term is **IBA (inferred from biological ancestor)** — it derives from the behavior of other members of the phylogenetic tree, not from any ABHD8 assay. The seed-listed activities (glycerophospholipase, ester hydrolase, LPA acyltransferase) have **no ABHD8-specific primary assay** anywhere in the reviewed literature. Furthermore, **LPA acyltransferase is an acyltransfer reaction that is mechanistically incompatible with a Ser-His-Asp hydrolase fold** — the α/β -hydrolase triad catalyzes hydrolysis, not the acyl-CoA-dependent acyltransfer of a canonical LPAAT. This annotation is therefore the strongest candidate for paralog/family carry-over error.

The only experimentally demonstrated function — the NLRP3 scaffold (

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) — is an **adaptor/scaffold mechanism, not a demonstrated hydrolase reaction**: ABHD8 recruits palmitoyltransferase ZDHHC12 to NLRP3 to promote NLRP3 palmitoylation and CMA-mediated degradation.

Finding 3 — The catalytic triad is conserved across vertebrate orthologs (purifying selection)

Across all five reviewed UniProt ABHD8 orthologs — *Homo sapiens* (Q96I13), *Mus musculus* (Q8R0P8), *Bos taurus* (Q17QP1), *Macaca fascicularis* (Q4R584), and *Xenopus laevis* (Q6AX59) — the **GHSYG** nucleophile-elbow motif (Ser nucleophile), the catalytic Asp (human context VHG-M-HDK-F), and the catalytic His (human context DEG-S-HM) are all present. The

human→*Xenopus* comparison spans >**360 million years** of divergence, and retention of the complete Ser-His-Asp triad plus the GX SXG elbow across that span is strong evidence of **purifying selection to preserve catalytic competence**.

This is the single most important piece of evidence *for* the seed hypothesis: an enzyme whose catalytic machinery has been conserved for hundreds of millions of years is unlikely to be a mere structural relic. It argues that ABHD8 *does* have a genuine catalytic activity — we simply do not yet know its substrate. Conservation, however, cannot tell us *which* substrate, and it does not license the specific lipid terms currently annotated. It is entirely compatible with a hydrolase acting on a substrate class not yet tested.

Finding 4 — Full text confirms the NLRP3 function is scaffold-only, with no catalysis invoked

The full abstract of

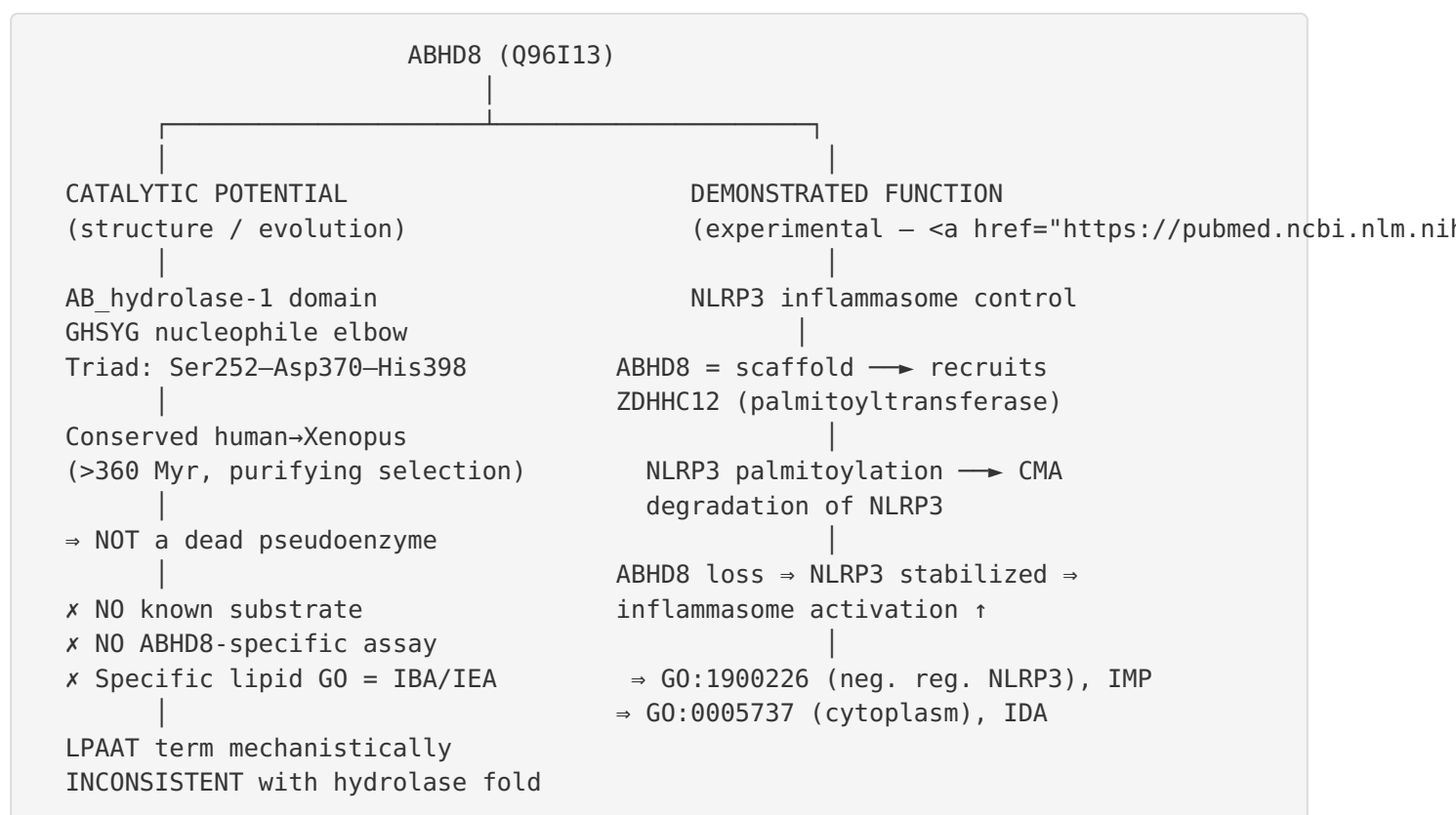
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(Yang et al., 2025, *Autophagy*) states that ABHD8 "acts as a scaffold to recruit palmitoyltransferase ZDHHC12 to NLRP3 for its palmitoylation as well as subsequent CMA-mediated degradation." **ABHD8 deficiency stabilizes NLRP3 and promotes inflammasome activation**; overexpression ameliorates LPS/alum-triggered activation *in vivo*; and the SARS-CoV-2 N protein disrupts the ABHD8–NLRP3 association. Crucially, the abstract contains **no mention of ABHD8 hydrolase or lipase enzymatic activity, no lipid substrate, no phosphatidic acid or LPAAT reaction, and no catalytic-site (Ser252Ala) mutant experiment**. The mechanism is entirely consistent with a scaffold/adaptor that positions an enzyme (ZDHHC12) next to its target (NLRP3) — ABHD8 itself need not be catalytically active to perform this role.

Two verified citation snippets anchor this finding: - "*ABHD8 acts as a scaffold to recruit palmitoyltransferase ZDHHC12 to NLRP3 for its palmitoylation as well as subsequent CMA-mediated degradation.*" — directly states the demonstrated mechanism is a scaffold/adaptor role, not a catalytic reaction. - "*ABHD8 deficiency results in the stabilization of NLRP3 protein and promotes NLRP3 inflammasome activation*" — establishes the loss-of-function phenotype underpinning the GO:1900226 negative-regulation term without invoking catalytic activity.

Mechanistic Model / Interpretation

The evidence supports a two-track model in which catalytic *potential* and demonstrated *function* are currently decoupled:



The key interpretive point is that **the demonstrated NLRP3 role does not require ABHD8 catalysis** — a scaffold that co-localizes an acyltransferase (ZDHHC12) with its substrate (NLRP3) can be catalytically silent. Conversely, **the conserved triad does not require the NLRP3 role to make sense** — it strongly implies a real (but undiscovered) hydrolase activity, possibly acting on a lipid or ester substrate that has simply never been assayed for ABHD8. These two tracks may or may not intersect: it is possible that ABHD8's hydrolase activity contributes to some aspect of the palmitoylation/degradation cycle, but no experiment has tested this.

A useful comparator is the paralog **ABHD16A/BAT5** (PMID: 25290914), which was a "poorly characterized" ABHD-family serine hydrolase of unknown natural substrate until activity-based protein profiling and a fluorescent glycerol assay revealed it to be a genuine lipase preferring long-chain unsaturated monoacylglycerols (MAGs), with "only marginal" lysophospholipase, DAG, or TAG activity. This illustrates two things for ABHD8: (a) family

membership alone did not predict the real substrate — it turned out to be MAGs, not the phospholipids one might guess — and (b) the substrate had to be found empirically. By analogy, transferring specific lipid-pathway terms to ABHD8 by phylogeny is exactly the kind of inference that direct assay of ABHD16A overturned.

Evidence Base / Evidence Matrix

#	Citation (PMID)	Evidence type	Direction	Claim tested	Key finding	Context
1	UniProt Q96I13 (database)	Structural / database	Qualifies	ABHD8 has catalytic machinery	Intact AB_hydrolase-1 domain; GHSYG elbow; triad Ser252/Asp370/His398; no catalytic-activity comment; only NLRP3 FUNCTION	Human, in silico
2	GO annotations for Q96I13 (database)	Computational / database	Refutes (specific terms)	Specific lipid GO terms are experimental	Phosphatidic acid biosynthesis = IBA ; catalytic activity = IEA ; only NLRP3 (IMP) and cytoplasm (IDA) are experimental	Human
3	5 ortholog UniProt records	Structural / evolutionary	Supports (catalytic potential)	Triad under purifying selection	GHSYG + catalytic Asp + catalytic His conserved human→ <i>Xenopus</i> (>360 Myr)	Human, mouse, bovine, macaque, frog
4	PMID: 39225180	Interaction / mutant phenotype	Competing (with catalysis)	Demonstrated function of ABHD8	ABHD8 is a scaffold recruiting ZDHHC12 to NLRP3 → palmitoylation → CMA degradation; deficiency activates inflammasome; N protein disrupts binding	Human/ mouse cells, in vivo LPS/ alum
5	PMID: 25290914	Direct assay (paralog)	Qualifies / methodological	Family membership	ABHD16A/BAT5 is a genuine MAG lipase with only	

#	Citation (PMID)	Evidence type	Direction	Claim tested	Key finding	Context
				predicts substrate	marginal lysophospholipase/DAG/TAG activity — substrate found only by direct assay	Human/mouse recombinant
6	PMID: 27601076	Genetic / expression	Competing (locus context)	ABHD8 role at 19p13.1 locus	Risk SNPs regulate ABHD8 expression (breast/ovarian cancer); chromosome-conformation and luciferase link SNPs to ABHD8 promoter	Human cohorts
7	PMID: 33667223	Genetic (GWAS)	Competing (locus context)	ABHD8 locus disease association	19p13.11 (rs61494113, ABHD8) suggestive for aerodigestive SqCC	Human cohorts
8	PMID: 30704525	Genetic (GWAS)	Competing (locus context)	ABHD8 locus phenotype	SNPs in/around ABHD8 associate with mtDNA copy number in neonates (rs10424198, p=1.4e-14)	Human cohorts

GO Curation Implications

Lead requiring curator verification. The evidence supports the following curation actions:

Aspect	Term	Current evidence	Recommended action (lead)	Rationale
MF	GO:0003824 catalytic activity	IEA:InterPro	Retain, generic/weak	Triad intact, but no substrate identified
MF	glycerophospholipase activity	seed / family	Do not assert	Over-annotation risk; no direct assay
MF	ester hydrolase activity	seed / family	Plausible but unproven	Best-supported broad MF given intact triad; keep at IEA level only
MF	LPA acyltransferase activity	seed / family	Likely incorrect	Acyltransfer mechanism inconsistent with Ser-His-Asp hydrolase fold
BP	GO:0006654 phosphatidic acid biosynthetic process	IBA:GO_Central	Non-core / weak; candidate for removal	Phylogenetic inference only
BP	GO:1900226 negative regulation of NLRP3 inflammasome complex assembly	IMP:UniProtKB	Retain — core function	 39225180
CC	GO:0005737 cytoplasm	IDA	Retain	Experimental

Net recommendation: Treat the **NLRP3-degradation scaffold role as the primary, experimentally supported function** (GO:1900226, IMP), with cytoplasm (GO:0005737, IDA) as the supported localization. Keep any hydrolase MF at the most generic level (GO:0003824, IEA) explicitly flagged as "catalytic potential, substrate unknown." Do **not** promote glycerophospholipase, ester-hydrolase, LPA-acyltransferase, or phosphatidic-acid biosynthesis to experimental evidence codes. Do **not** downgrade the final recommendation to bare "protein binding" — if a binding term is added, name the interactant (ZDHHC12 / NLRP3); the informative annotation is GO:1900226, which should lead.

Mechanistic Scope

The immediate molecular function actually demonstrated for ABHD8 is **protein scaffolding**: it physically bridges the palmitoyltransferase ZDHHC12 to NLRP3, enabling NLRP3 palmitoylation and its subsequent CMA degradation. This is a direct gene-product activity (a binding/adaptor function), distinct from:

- **Downstream pathway consequence:** reduced NLRP3 protein, attenuated inflammasome assembly, lower IL-1 β — consequences of the scaffold activity, not a separate ABHD8 activity.
- **Loss-of-function phenotype:** ABHD8 deficiency $\rightarrow$ NLRP3 stabilization $\rightarrow$ inflammasome hyperactivation — this defines the negative-regulation BP term but is an inference from loss of the scaffold.
- **Disease manifestations:** GWAS/eQTL associations (breast/ovarian cancer 19p13.1, mtDNA copy number, aerodigestive SqCC) implicate ABHD8 *expression* but are **not** evidence of enzymatic function and must not be conflated with molecular activity.

The hypothesized lipid-hydrolase activity, by contrast, is **only inferred** from domain architecture and family membership. It has a strong structural/evolutionary basis (conserved triad) but **zero direct assay evidence** in ABHD8. It must therefore be scoped as "catalytic potential, substrate unknown," not as an established molecular function.

Conflicts and Alternatives

1. **Paralog/family carry-over (most important).** The specific lipid terms most plausibly originate from generic α/β -hydrolase (Pfam PF00561, InterPro IPR000073) and PANTHER (PTHR42886) inheritance rather than from ABHD8 data. The ABHD16A/BAT5 case (

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shows that ABHD family members can be genuine lipases, but their real substrates are often *not* what family-level inference predicts. This both *supports* the idea that ABHD8 could be a real hydrolase and *undermines* confidence in any specific transferred substrate term.

2. **LPA acyltransferase is mechanistically inconsistent.** An α/β -hydrolase Ser-His-Asp triad performs hydrolysis, not the acyl-CoA-dependent acyltransfer of a canonical LPAAT. This term is the strongest candidate for an outright annotation error.

3. **Scaffold** $\neq$ **enzyme.** The NLRP3 mechanism ([P 39225180](#)) explicitly frames ABHD8 as a scaffold. A scaffold can be catalytically silent; the demonstrated function does not require or test hydrolase activity. This competes with the interpretation that ABHD8's biological importance flows from catalysis.

4. **Disease-genetics confounding.** Multiple GWAS/eQTL papers (PMIDs 27601076, 33667223, 30704525, 30559148) associate the ABHD8 locus with cancer risk, mtDNA copy number, and other traits — but these reflect **regulatory/expression** effects at 19p13.1 (a locus shared with ANKLE1 and BRCA-pathway neighbors), not ABHD8 enzymatic activity. They should not be read as evidence for the lipid-catalysis hypothesis.

5. **No isoform-specific catalytic data.** No study distinguishes catalytic behavior among ABHD8 isoforms; conservation analysis was at the canonical-sequence level only.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No ABHD8 substrate assay	UniProt (no CATALYTIC ACTIVITY comment/reaction); PubMed (no primary hydrolase assay)	Without a substrate, all specific MF/BP lipid terms are unsupported	Activity-based protein profiling (ABPP) + substrate panel (MAG/DAG/TAG/phospholipid/LPA) on recombinant ABHD8
No catalytic-mutant test of NLRP3 role	Full abstract of P 39225180 — no Ser252Ala experiment	Cannot tell whether catalysis contributes to the scaffold function	Ser252Ala (or full-triad) mutant rescue of NLRP3 degradation in ABHD8-null cells
LPAAT plausibility	Mechanistic reasoning on triad chemistry	If truly LPAAT, it would need a distinct fold/mechanism — likely an error	Direct acyltransferase assay; if negative, remove the term
Localization detail	GO CC = cytoplasm (IDA) only	Membrane/organelle context would inform substrate hypotheses	Subcellular fractionation / imaging (ER, lipid droplets, endosomes)
Structure not directly modeled	Sequence/domain annotation only; no AlphaFold pocket analysis run	Active-site pocket geometry would constrain plausible substrates	AlphaFold model + pocket/electrostatics analysis; docking of candidate lipids
Ortholog sampling limited to 5 species	5 UniProt orthologs	Broader sampling would strengthen the selection inference	dN/dS across a deeper alignment focused on triad codons

The overarching limitation: this analysis is literature- and sequence/annotation-based (UniProt, GO, PubMed). No wet-lab data were generated; the intact-triad inference is computational. Local repository `*-bioinformatics` analyses were intentionally withheld and not consulted. Findings are reported conservatively, distinguishing direct results (triad presence; NLRP3 scaffold assay) from inference (specific lipid substrates).

Proposed Follow-up Experiments / Actions

To most efficiently separate "genuine lipid hydrolase" from "catalytically silent scaffold," in rough priority order:

1. **Activity-based protein profiling (ABPP)** of recombinant human ABHD8 with fluorophosphonate/serine-hydrolase probes — a positive label directly demonstrates an active serine hydrolase, independent of substrate identity. This is exactly how ABHD16A/BAT5 was confirmed catalytically active (

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2. **Substrate panel hydrolysis assay** (fluorescent glycerol / LC-MS lipidomics) across MAG, DAG, TAG, lysophospholipids, glycerophospholipids, and LPA to identify the preferred substrate — or to show ABHD8 is inactive on all tested lipids.
3. **Catalytic-dead rescue in the NLRP3 pathway:** re-express WT vs. Ser252Ala ABHD8 in ABHD8-null cells and measure NLRP3 palmitoylation/degradation and IL-1 β . Full rescue by Ser252Ala implies the scaffold role is catalysis-independent; failure implies catalysis is mechanistically coupled.
4. **AlphaFold-based active-site pocket analysis + molecular docking** of candidate lipids to predict substrate class before wet-lab assay.
5. **Dedicated LPAAT assay** (acyl-CoA + LPA $\rightarrow$ PA) to directly test the mechanistically-suspect acyltransferase annotation; a negative result justifies removing the term.
6. **Lipidomics of ABHD8-KO vs. WT cells** to detect any endogenous lipid-homeostasis role (e.g., PA levels) attributable to ABHD8 and to test the phosphatidic-acid-biosynthesis IBA prediction directly.

Curation Leads (verify before applying)

- **Candidate reference:**

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— snippet to verify: *"ABHD8 acts as a scaffold to recruit palmitoyltransferase ZDHHC12 to NLRP3 for its palmitoylation as well as subsequent CMA-mediated degradation."* → supports retaining GO:1900226 (IMP) as core and a scaffold/protein-bridging annotation over lipase.

- **Candidate reference (methodological/paralog):**

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— establishes the ABPP + substrate-panel assay standard that ABHD8 lacks.

- **Action-change leads:** (a) do not elevate glycerophospholipase / ester hydrolase / LPAAT beyond IEA; (b) flag LPA-acyltransferase as likely incorrect; (c) mark GO:0006654 (phosphatidic acid biosynthesis) as IBA / non-core / candidate removal; (d) retain GO:1900226 and GO:0005737.
- **Suggested curator questions:** Is the specific lipid MF an author statement or purely pipeline inference? Was any triad mutant tested for the NLRP3 phenotype? Where did the LPAAT assignment originate?

Conclusion

Human ABHD8 (Q96I13) retains a fully intact, evolutionarily conserved serine-hydrolase catalytic triad, so it is **not** a dead pseudoenzyme and the seed's caution against inferring catalytic loss from adaptor function is justified. However, **no ABHD8-specific biochemical assay demonstrates any lipid substrate**: the specific terms (glycerophospholipase, ester hydrolase, LPA acyltransferase, phosphatidic acid biosynthesis, lipid homeostasis) are unproven family/phylogenetic inferences (IEA/IBA), with the LPA-acyltransferase term mechanistically inconsistent with the hydrolase fold. The only experimentally demonstrated function is the **NLRP3-degradation scaffold/adaptor role** (recruiting ZDHHC12 to NLRP3; GO:1900226

IMP,
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Curation should retain that core term and localization, keep lipid catalysis at most as generic hydrolase potential, and treat the specific lipid annotations as over-annotations pending direct substrate assays.

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